Exercise 3

SQL CASE Statements

① The products Table

Products

product_id	product_name	price
1	Laptop	1200.00
2	Phone	800.00
3	keyboard	45.00
4	Monitor	300.00
5	Mouse	25.00

① Classify each product by price into the three tier below.

```
SELECT
        product_name,
        price,
        CASE
    WHEN price > 1000 THEN 'Expensive'
    WHEN price BETWEEN 100 AND 1000
        THEN 'Mid_range'
    WHEN price < 100 THEN 'Budget'
    END As price_Category
FROM product;
```

product_name	price	price_Category
Laptop	1200.00	expensive
Phone	800.00	Mid_range
keyboard	45.00	Budget
Monitor	300.00	Mid_range
Mouse	25.00	Budget

② The Orders Table

Orders

order_id	customer_name	amount
1	Alice	150.00
2	Bob	560.00
3	Charlie	999.99
4	Diana	45.50
5	Ethan	1200.00

② Label each order by its value.

```
SELECT
        Customer_name,
        amount,
         CASE
        WHEN amount >1000 THEN 'High value'
        WHEN amount BETWEEN 500 AND 999.99
        THEN 'Medium value'
        WHEN amount < 500 THEN 'Low value'
    END As order_value_category
FROM orders;
```

customer_name	amount	order_value category
Alice	150.00	Low value
Bob	560.00	Medium value
Charlie	999.99	Medium value
Diana	45.50	Low value
Ethan	1200.00	High value

③ The employees Table

employees

emp_id	emp_name	department	Salary
1	John	IT	85000
2	Sara	HR	60000
3	Mark	IT	75000
4	Lucy	Finance	95000
5	Tom	HR	55000

③ Categorize each employee's position level high both department and Salary.

```
SELECT emp_name, department, Salary,
~~CASE~~ CASE
~~IF~~      WHEN department = 'IT' AND 'Salary' >
            80000 THEN Senior 'IT'
            WHEN department = 'HR' AND 'Salary' >
             55000 THEN 'Experienced HR'
            Else 'Staff'
            END AS position_level
FROM employees;
```

emp_name	department	Salary	position_level
John	IT	85000	Senior IT
Sara	HR	60000	experienced ~~expensive~~ HR
Mark	IT	75000	Staff
Lucy	Finance	95000	Staff
Tom	HR	55000	experienced HR

④ The students Table

Students

student_id	student_name	score
1	Anna	92
2	Ben	76
3	Cara	59
4	David	83
5	Ella	68

⑪ Assign each student a letter grade based on their score.

```
SELECT student_name, score, grade,
    CASE
        WHEN score ≥ 90 THEN 'A'
        WHEN score BETWEEN 80 AND 89 THEN 'B'
        WHEN score BETWEEN 70 AND 79 THEN 'C'
        WHEN score BETWEEN ≤ 60 THEN
        WHEN score BETWEEN 60 AND 69
        THEN 'D'
        WHEN score ≤ 60 THEN 'F'
    END AS grade
FROM students;
```

~~student_name~~

student_name	score	grade
Anna	92	A
Ben	76	C
Cara	59	F
David	83	B
Ella	68	D

⑤ The deliveries Table

deliveries

delivery_id	delivery_time_minutes
1	45
2	80
3	30
4	65
5	100

⑤ Label delivery performance based on the time taken.

```
SELECT deliver_id, deliy_time,
        CASE
    WHEN delivery_time ≤ 30 THEN 'Fast'
    WHEN delivery_time_minutes BETWEEN
     31 AND 60 THEN 'On Time'
     WHEN > 60 THEN 'Late'
     END As performance
FROM deliveriesj
```

delivery_id	delivery_time_minutes	performance
1	45	On_time
2	80	Late
3	30	Fast
4	65	Late
5	100	Late

≤ 30 mins → 'fast'
31-60 mins → 'On Time'
> 60 mins → 'Late'

⑥ The tickets Table

tickets

ticket_id	issue_type	priority
1	Login issue	1
2	Server down	3
3	Slow System	2
4	Email error	2
5	Password reset	1

⑥ Convert the numeric priority into a readable label.

```
SELECT issue_type, priority,
        CASE
    WHEN priority = 3 THEN 'High'
    WHEN priority = 2 THEN 'Medium'
    WHEN priority = 1 THEN 'Low'
  END AS priority_label
FROM tickets;
```

issue_type	priority	priority_label
Login issue	1	Low
Server down	3	High
Slow System	2	Medium
Email error	2	Medium
Password reset	1	Low

3 -> 'High'
2 -> 'Medium'
1 -> 'Low'

① The attendance Table

attendance

student_id	days_present	total days
1	45	50
2	30	50
3	48	50
4	25	50
5	50	50

② Calculate attendance percentage and classify the result

```
SELECT student_id,
       days_present/total_days *100
       As attendance_percentage,
       CASE
  WHEN days_present/total_days *100
  ≥ 90 THEN 'Excellent'
  WHEN days_present/total_days x100 BETWEEN 75
   AND 89 THEN 'Good'
  WHEN days_present/total_days x100 < 75
  THEN 'Need Improvement'
  END As attendance_status
FROM attendance;
```

student_id	attendance_percentage	attendance_status
1	90	Extellent
2	60	Need Improvement
3	96	Excellent
4	50	Need Improvement
5	100	Excellent

⑦ The product_inventory Table

products_inventory :

product_id	stock_qty
1	5
2	0
3	25
4	10
5	3

⑧ Label the stock status of each product.

```
SELECT product_id, stock_qty,
    CASE
    WHEN stock = 0 THEN 'Out of stock'
    WHEN stock BETWEEN 1 AND 5 THEN 'Low stock'
    WHEN stock > 5 THEN 'In stock'
  END AS stock_status
FROM products_inventory;
```

product_id	stock_qty	stock_status
1	5	Low stock
2	0	Out of stock
3	25	In stock
4	10	In stock
5	3	Low stock

0 -> 'out of stock'
1-5 -> 'Low stock'
> 5 -> 'In stock'

① The attendance Table

attendance

student_id	days_present	total_days
1	45	50
2	30	50
3	48	50
4	25	50
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  WHEN days_present/total_days x100 < 75
  THEN 'Need Improvement'
  END As attendance_status
FROM attendance;
```

student_id	attendance_percentage	attendance_status
1	90	Excellent
2	60	Need Improvement
3	96	Excellent
4	50	Need Improvement
5	100	Excellent

⑨ The Classes Table

Classes

class_id	subject	enrolled_students
1	Math	30
2	English	25
3	Science	15
4	Art	5
5	History	20

⑨ classify each class by the number of enrolled students.

```
SELECT subject, enrolled_students,
    CASE
        WHEN class ≥ 25 THEN 'Large'
        WHEN class BETWEEN 10 AND 24 THEN
        'Medium'
        WHEN class < 10 THEN 'Small'
    END AS class_size_category
FROM classes;
```

subject	enrolled_students	class_size_category
Math	30	Large
English	25	Large
Science	15	Medium
Art	5	Small
History	20	Medium

≥ 25 → 'Large'
10–24 → 'Medium'
< 10 → 'Small'

⑩ The payments Table

payments

payment_id	amount	payment_method
1		
2		
3		
4		
5		

⑩ Apply a discount flag based on the payment method and amount.

```
SELECT payment_id, payment_method, amount,
    CASE
        WHEN payment_method = 'cash' AND amount >= 200
        THEN 'Eligible for Discount'
        Else 'Non Eligible'
    END As discount_eligibility
FROM payments;
```

payment_id	payment_method	amount	discount_eligibility
1	Card	50.00	Not Eligible
2	Cash	200 00	Eligible for Discount
3	Card	150.00	Not Eligible
4	Paypal	75 00	Not Eligible
5	Cash	300.00	Eligible for Discount